Last time: talked about how to find a particular solution denoted y_p to $ay'' + by' + cy = f(t)$ for a few kinds of functions $f(t)$ as follows:

(1) If $f(t)$ is a polynomial, usually try y_p to be a polynomial of the same degree.

(2) For k constant, if $f(t) = e^{kt}$, usually try
$y_p = Ae^{kt}$

(3) If $f(t) = \cos(kt)$ or $\sin(kt)$, usually try
$y_p = A\cos(kt) + B\sin(kt)$

But in all of these, we have these unknown constants A, B, etc., we have to explicitly find those in a specific problem. How to do that?

Well differentiate this y_p the appropriate number of times and plug in. Equate coefficients and solve. Will give a linear system (linear algebra ideas again!)

in the unknowns that you are trying to find. This is called the <u>method of undetermined coefficients (MUC).</u>

And recall for sum/difference/product of such terms can still do this by combining the proposed guess with same operation used to create the combination you are trying to equal.

So really three things involved:

(1) Is undetermined coefficients a viable approach for the problem in question?
(2) If so, what is the proper form to try for y_p?
(3) Solve for the appropriate coefficients in y_p

B4.4.3

$$2y'' - 6y' + y = \frac{\sin x}{e^{4x}} = e^{-4x} \sin x$$

Undetermined coefficients is viable due to latter form. Must be careful here as it would be easy to think it's not due to division as usually that's not allowed.

And indeed it wouldn't be if it was instead $\dfrac{e^{4x}}{\sin x}$

as $\csc x$ is not a viable function for undetermined coefficients.

So since it is viable what is our attempt? Product of the individual attempts:

$$y_p = e^{-4x}(A\cos x + B\sin x)$$

Seems natural as latter is usual thing to try for $\sin x$. And former is usual thing to try for e^{-4x}, the only thing which might concern us is why no constant in front as usual?

Well we could put constant C in front. But when you multiply it out, say the first term will have coefficient AC, an arbitrary constant. But given it's just one constant, why would we need two unknowns A and C to represent it?

Simple answer is we don't! The C is unnecessary/redundant so no point in having it.

This is actually quite important. Questions on undetermined coefficients can be quite long, particularly if there is product involved as there is here.

Taking second derivative of the proposed y_p and setting it all up can be quite long. You don't want to make it worse by including an unneeded constant! Important to be as efficient as possible.

B4.4.7

$$8z'' - 2z' = 3x^{100} e^{4x} \cos(25x)$$

This is product of three styles which are all viable so product is viable (via undetermined coefficients)
Usual try would be:

$$z_p = q(x)\, e^{4x} [A \cos(25x) + B \sin(25x)]$$

degree 100

B4.4.11

$$y'' + y = 2^x = e^{x \ln 2}$$

At first glance might not look like undetermined coefficients is viable, but rewriting it, clearly is. So try:

$$y_p = Ae^{x \ln 2} = A \cdot 2^x$$

$$y_p' = A(\ln 2)e^{x \ln 2} = A(\ln 2)2^x$$

$$y_p'' = A(\ln 2)^2 e^{x \ln 2} = A(\ln 2)^2 2^x$$

Plugging in, differential equation becomes

$$A \cdot 2^x[(\ln 2)^2 + 1] \stackrel{?}{=} 2^x$$

Equating coefficients yields:

$$A = \frac{1}{(\ln 2)^2 + 1}$$

Important nuance in undetermined coefficients. When proposing what y_p would work for which $f(t)$, I was careful to write/say "usually" try such and such.
And that's because there are exceptions as we now will observe. Say we want a particular solution for:

$$y'' + y = \cos t$$

The instinctive try is:

$$y_p = A \cos t + B \sin t$$

From this:

$$y_p'' = -A \cos t - B \sin t$$

Meaning we have

$$y_p'' + y_p = 0$$

That is, that sum is identically 0. But we want it to equal $\cos t$ which of course would now be impossible.

Hence our choice of what y_p ought to be must have been incorrect. But why? And what is correct?

Key is $\cos t$ is solution to homogeneous counterpart of this equation, i.e. $y'' + y = 0$. That's why it can't logically work in the nonhomogeneous equation.

That is, there is *overlap* between the homogeneous solution and what we are trying to equal in the nonhomogeneous equation.

How to fix? Multiply usual guess by t. Why? Similar idea as to why when we had repeated root in characteristic equation of homogeneous equation we multiplied the usual solution by t, based on linear algebra ideas that might be clarified a bit later.

If the root of characteristic equation which causes the overlap is a double root of that, then multiply usual guess by t^2 instead, and beyond if necessary if repeated more than twice (higher order equation)

Find particular solution of

$$y'' + y = t \cos t$$

What to try? Again initial instinct would be

$$y_p = (At + B)\cos t + (Dt + E)\sin t$$

How do we know this is not logical though, without plugging in and seeing it doesn't work? We don't want to do that then realize it doesn't work, that would be a big waste of time!

Key is the $B \cos t$ by itself will reduce to 0 on left side so there won't be B after reduction. In essence that term would be irrelevant so doesn't make sense to include that in the proposed try (same for $E \sin t$).

Whether the rest of it works or not (it doesn't) that's why that form wouldn't be a logical overall attempt.

Correct is to multiply that usual attempt by t due to the overlap, i.e.

$$y_p = (At^2 + Bt)\cos t + (Dt^2 + Et)\sin t$$

Don't write $(At^2 + Bt + C)\cos t$ for the first term. That's a tempting thing to do to make it the "full" second second degree polynomial multiplying.

But again the $C \cos t$ term won't contribute anything since that is a solution to the homogeneous equation.

So while including that won't make it wrong, it also won't do anything, will just make it worse in adding more to be done in the calculation. And this calculation will be bad enough as is! Don't make it worse for no reason!

What particular solution to try for

$$y'' - 2y' + y = e^t$$

Characteristic equation of homogeneous counterpart is

$$r^2 - 2r + 1 = 0$$

----> $r = 1, 1$ ----> double overlap

So should try

$$y_p = At^2 e^t$$

Then $y_p' = (At^2 + 2At)e^t$

----> $y_p'' = (At^2 + 4At + 2A)e^t$

Plugging all these into equation gives:

$$(At^2 + 4At + 2A)e^t - (2At^2 + 2At)e^t + At^2 e^t = e^t$$

----> $2Ae^t = e^t$

----> $A = \frac{1}{2}$

Again don't include Bte^t or Ce^t in the prospective guess as those won't contribute anything and will just make the calculation longer.

B4.4.25

$$y'' + 2y' + 4y = 111e^{2t}\cos(3t)$$

Want particular solution. But we are maybe unsure what form to try since at start we are uncertain if there is overlap with the homogeneous solution.

So find solution to corresponding homogeneous equation first. In fact that's what you should pretty much always do with any nonhomogeneous equation.

Again you don't want to do the full calculation with the wrong form only to realize at the end that it was so. That will waste a lot of time!

Characteristic equation of homogeneous counterpart is

$$r^2 + 2r + 4 = 0$$

----> $$(r+1)^2 + 3 = 0$$

----> $r = -1 \pm \sqrt{3}i$

Thus

$$y_c = c_1 e^{-t} \cos(\sqrt{3}\,t) + c_2 e^{-t} \sin(\sqrt{3}\,t)$$

Definitely no overlap here, nothing like what you are trying to equal.

So would use the usual try:

$$y_p = Ae^{2t} \cos(3t) + Be^{2t} \sin(3t)$$

B4.4.27

$$y'' + 9y = 4t^2 \sin(3t)$$

Again solve homogeneous counterpart (complementary equation) first.

Characteristic equation of that is:

$$r^2 + 9 = 0$$

----> $$r = \pm 3i$$

Meaning we have:

$$y_p = c_1 \cos(3t) + c_2 \sin(3t)$$

Seems to have same type of term as the solution we are trying to create in the nonhomogeneous equation. Thus we multiply the usual guess by t meaning here we try:

$$y_p = (At^3 + Bt^2 + Ct)\cos(3t) + (Et^3 + Ft^2 + Gt)\sin(3t)$$

Again don't include D cos (3t) style term. Not wrong if you do, but again that term will contribute nothing since it is a solution to the complementary equation. So again you will just make it harder on yourself for no reason.

Ex: Another example to contrast a bit the nuances of "overlap"

$$y'' + 9y = e^{t} \sin(3t)$$

Find form of prospective particular solution. From last question:

Characteristic equation of that is:

$$r^2 + 9 = 0$$

----> $$r = \pm 3i$$

Meaning we have:

$$y_c = c_1 \cos(3t) + c_2 \sin(3t)$$

Is there overlap between this and the solution you are trying to create, i.e. $e^{t} \sin(3t)$?

This can be tricky, must be careful. Some would argue no because this has exponential term and our y_c does not. Others would say yes since both have $\sin(3t)$. But which is correct?

Those saying no, earlier we claimed $t \cos t$ did overlap with $\cos t$, even though former had linear term, and latter did not. And in prior question we did say that $4t^2 \sin(3t)$ overlaps with this same complementary solution that we have in this example.

So what is the difference if answer here is that they don't overlap?

Key is: if ratio is polynomial, that's when there is overlap.

Hence why $t \cos t$ and $\cos t$ do overlap as do $\sin(3t)$ and $4t^2 \sin(3t)$.

But $\sin(3t)$ and $e^t \sin(3t)$ do not since their ratio is

an exponential function not a polynomial, that's the key.

Why? Well relates to same linear algebra idea that causes you to multiply by t for second solution when you had repeated root in characteristic equation of homogeneous equation.

Also same type of idea that causes you to multiply by t when you have overlap between homogeneous solution and function you are trying to create in nonhomogeneous equation. Again will clarify that a bit later in course when we put it all together.

So since no overlap here, would do the usual:

$$y_p = Ae^t \cos(3t) + Be^t \sin(3t)$$

B4.5 Superposition Principle/Clarification of MUC

Last section we worried about finding particular solution to nonhomogeneous equation. Now how to find general solution?

Already alluded to this and fortunately same idea that we often used in solving nonhomogeneous matrix equations, $Ax = b$

That is, find general solution to the complementary equation (i.e. the homogeneous counterpart) and add that to a particular solution. A generalization of that is what we call the <u>superposition principle:</u>

Say y_1 is a solution to

$$ay'' + by' + cy = f_1(t)$$

and y_2 is a solution to

$$ay'' + by' + cy = f_2(t)$$

Then for any constants k_1, k_2, we have $k_1 y_1 + k_2 y_2$ is a solution to:

$$ay'' + by' + cy = k_1 f_1(t) + k_2 f_2(t)$$

This is natural based up on the fact that derivative behaves well under addition and scalar multiplication.

And in the special case when $k_1 = 1$ and $f_2(t) = 0$ we have y_2 as a solution to the complementary equation.

So overall this is how we find general solution to the nonhomogeneous equation, the $k_2 y_2$ is the extra thing which generalizes the solution.

To make it clear: for

$$ay'' + by' + cy = f(t)$$

If y_c is the general solution to:

$$ay'' + by' + cy = 0$$

and y_p is a particular solution to, then general solution to that nonhomogeneous equation is:

$$y(t) = y_c + y_p$$

Like with matrix equations, any solution is of this type since the difference of two solutions to the nonhomogeneous equation is a solution to the homogeneous equation, hence why this is the general solution.